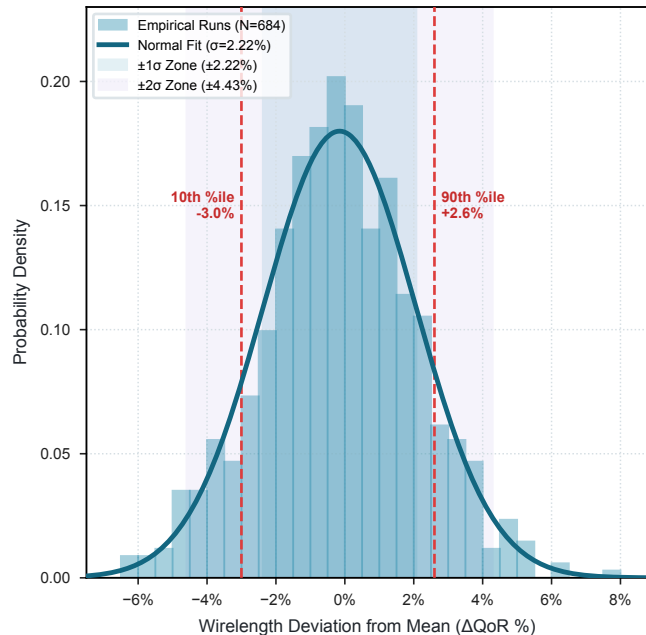
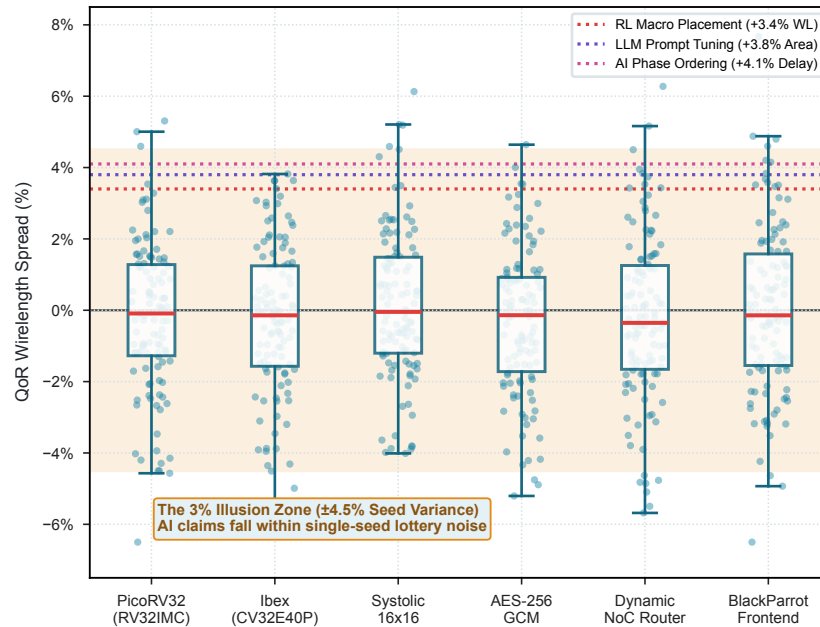


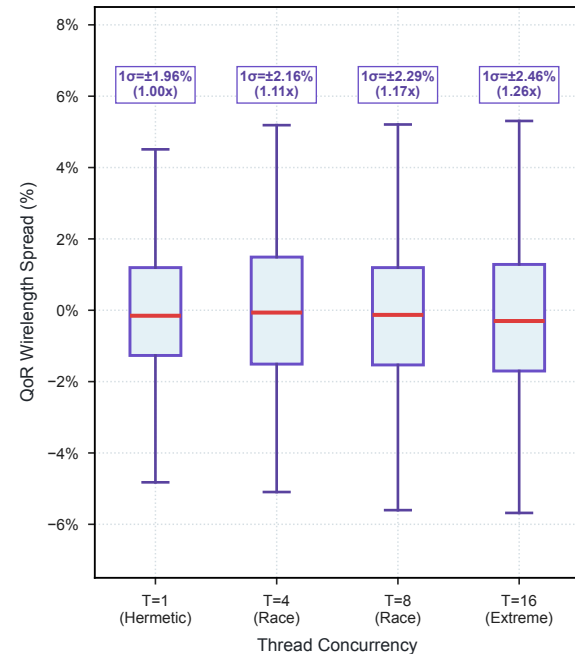
A. QoR Dispersion (N=684 Signoff Runs)



B. 'The 3% Illusion': AI Claims vs. Empirical Seed Lottery



C. Thread Race Concurrency



EMPIRICAL TAKEAWAY: Physical EDA flows (OpenROAD/Yosys/OpenSTA) exhibit a natural 3%–8% QoR dispersion ($1\sigma = \pm 2.22\%$, peak-to-peak = 14.2%) on identical RTL across seeds and thread races. Published AI-for-EDA gains (+3% to +5%) evaluated on single seeds measure stochastic noise rather than true optimization.